VASTRA
KOMPRESSOR AB - SWEDEN

Oil-injected rotary screw compressor, 75 kW

VRS-75

Service Manual

Translation of the original instructions (Swedish: Servicehandbok)

Document no.	**9821 0075 01**
Revision	**B (2024-02-12)**
Valid for	**VRS-75, serial no. from VRS75-200001**
Controller	**Vastra Vcon 3, firmware 4.x**
Replaces	**Rev. A, 2021-09**

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About this manual

This service manual is intended for service technicians. It describes the maintenance, the fault finding and the spare parts of the VRS-75. Installation and first start are described in the separate Installation Instructions 9821 0075 03. Keep the manual available at the compressor for the full lifetime of the product.

1 Safety

1.1 Safety symbols

The symbols below are used in this manual and on the labels on the compressor. Damaged or missing labels shall be replaced without delay. Label sets are ordered from Vastra service; state the serial number of the compressor.

General danger. Read the text that belongs to the symbol.

Dangerous electrical voltage. Electrical cabinet, motor terminal box.

Hot surface. Element outlet, discharge pipe, separator vessel and oil cooler can reach more than 110 C.

Rotating parts. Cooling fan and coupling. The fan can also run when the compressor is stopped.

System under pressure. The separator vessel stays pressurised after stop until it is vented.

Automatic start. The compressor can start without warning in automatic mode and after a power failure.

Read the manual before you start the work.

Wear hearing protection when you work at a running compressor with the doors open.

Wear safety glasses and protective gloves when you handle hot oil and when you vent the compressor.

Open fire and smoking are not allowed near the oil system and the separator vessel.

1.2 Signal words

DANGER
Indicates a hazardous situation that, if not avoided, will result in death or serious injury.

WARNING
Indicates a hazardous situation that, if not avoided, could result in death or serious injury.

CAUTION
Indicates a situation that, if not avoided, could result in minor injury or damage to the compressor.

NOTE
Gives information that is important for correct maintenance. Observe the note.

1.3 General safety precautions

- The operating company is responsible for that the compressor is operated and maintained only by competent persons who have read this manual.
- It is not allowed to make changes on the compressor or its control system without written approval from Vastra Kompressor AB.
- The compressed air is not suitable for breathing unless it is purified according to EN 12021.
- Never run the compressor above the maximum working pressure on the data plate, and never block, adjust or remove the safety valve.

- Never run the compressor with the doors or panels open, except for the short checks described in this manual.
- Do not use the emergency stop for normal stopping. The emergency stop does not vent the separator vessel.

1.4 Before maintenance and repair

Carry out the following steps before every maintenance or repair task, if the task description does not say otherwise.

1. Stop the compressor with the stop key (O). Wait until the unloaded run-down is finished and the motor has stopped.
2. Switch off the main isolator and lock it in the off position. Hang a sign "Work in progress - do not start" on the isolator.
3. Close the air outlet valve to the network. Open the manual vent valve on the separator vessel and control on the display (PT-1) that the internal pressure is 0 bar.
4. Let the compressor cool down. The element and the oil stay hot for a long time after stop.

WARNING

Observe that the stop key alone does not make the compressor safe. In automatic mode, and after a power failure with auto restart activated, the compressor starts without warning. Always lock the main isolator.

1.5 Personnel

Maintenance and fault finding as described in chapters 4 and 5 may be carried out by service technicians who are trained on Vastra VRS compressors. Work in the electrical cabinet and on the motor requires an electrically qualified person according to the national regulations. Repairs on the compression element are reserved for Vastra service.

2 General description

2.1 Introduction

The VRS-75 is an oil-injected, single-stage rotary screw compressor. The compression element is driven by a 75 kW electric motor through a flexible coupling. The compressor is air-cooled and is delivered as a complete package in a sound-insulated enclosure, with the Vastra Vcon 3 controller. Capacity regulation is by load and unload with the inlet valve; the motor runs at constant speed. For variable speed regulation, see the VRS-110.

2.2 Air flow and oil system

Air is drawn in through the air filter (AF) and the inlet valve (IV) into the compression element (CE). Oil is injected into the element for cooling, sealing and lubrication. The mixture of air and oil leaves the element through the discharge pipe, where the discharge temperature sensor TS-2 is mounted, and enters the separator vessel (SV). Most of the oil is separated in the vessel by the change of direction. The remaining oil is removed by the separator element (OS). The air then passes the minimum pressure valve (MPV) and the aftercooler (AC) and leaves the compressor at the air outlet valve.

The internal pressure presses the oil from the separator vessel through the thermostatic valve (TV), the oil cooler (OC) and the oil filter (OF) back to the element. The thermostatic valve begins to open at 71 C; below this temperature the oil bypasses the cooler so that the compressor

reaches operating temperature quickly. Oil collected in the separator element is returned to the element through the scavenge line, which has a restrictor and a sight glass.

The minimum pressure valve keeps the internal pressure at minimum 4 bar during loaded operation, also when the network is empty, to assure the oil circulation. Condensate from the aftercooler is removed by the electronic water drain (EWD).

2.3 Sensors and switches

Designation	Function	Location
PT-1	Internal pressure	Separator vessel, top cover
PT-2	Net pressure	Air outlet, after the minimum pressure valve
TS-1	Air inlet temperature (ambient)	Air filter housing
TS-2	Discharge temperature (element outlet)	Discharge pipe, approx. 150 mm after the element outlet flange
PS-3	Oil filter differential pressure switch	Oil filter head
PS-4	Air filter vacuum switch	Air filter housing, outlet side
B21	Motor PTC thermistors	Motor winding, connected to terminal X3

2.4 Technical data

Parameter	Value
Pressure version	7.5 bar / 10 bar / 13 bar
Max. working pressure	7.5 / 10 / 13 bar(e)
Free air delivery (ISO 1217, Annex C)	13.1 / 11.4 / 9.6 m3/min
Shaft power at max. working pressure	74.0 / 75.2 / 74.8 kW
Motor	75 kW, IE3, 2-pole, IP55
Supply	400 V 50 Hz / 460 V 60 Hz, 3-phase
Rated motor current	134 A at 400 V / 116 A at 460 V
Oil capacity (system)	38 l
Recommended oil	Vastra SynLube 46 (synthetic) or mineral compressor oil ISO VG 46 (DIN 51506 VDL)
Discharge temperature, normal operation	75 - 95 C
Thermostatic valve, opening temperature	71 C
Permitted ambient temperature	+1 to +46 C
Air outlet temperature	Ambient + 10 C (approx.)
Cooling air flow	7.6 m3/s
Sound pressure level (ISO 2151)	71 dB(A)
Air outlet connection	G 2 1/2
Dimensions L x W x H	2,050 x 1,100 x 1,750 mm
Weight	1,480 kg

Where three values are given, they apply to the 7.5, 10 and 13 bar versions. The data plate is on the inside of the electrical cabinet door.

3 Vcon 3 controller

3.1 Control panel

The Vcon 3 has a graphical display, the keys Start (I), Stop (O), Reset, arrow up and down, Enter and Esc, and four LEDs: voltage on (green), automatic operation (green), warning (yellow) and shutdown (red, flashing). The main screen shows the net pressure, the discharge temperature, the operating status (stopped, loaded, unloaded, waiting for restart) and the running hours.

3.2 Menu structure

Menu	Content	Access
Measurements	PT-1, PT-2, TS-1, TS-2, separator dP (PT-1 minus PT-2), motor starts last hour	All users
Counters	Running hours, loaded hours, motor starts, time to next service per task	All users
Event log	Last 100 shutdowns and warnings with time, code and measured values	All users
Settings	Load and unload pressure, auto restart, local or remote control, date and time	Customer code
Service	Service intervals, counter reset, sensor calibration, factory settings	Service code

The service code is given to trained service technicians only. Settings changed in the service menu are written to the event log with the code E-00 "parameter changed".

3.3 Shutdowns and warnings

A shutdown stops the compressor directly: the motor is switched off, the inlet valve closes and the separator vessel is vented through the blow-off valve. The red LED flashes and the code is shown on the display. When the cause is removed, press Reset. The compressor can then be started again with the Start key, or starts by itself if it is in automatic mode.

A warning lights the yellow LED. The compressor continues to run. Warnings disappear by themselves when the condition is no longer present, except the service warning E-19, which is reset in the service menu when the maintenance is done.

3.4 Service interval counters

The controller counts the running hours down to the next maintenance for each task: oil, oil filter, air filter, separator element and general inspection. When a counter reaches zero, warning E-19 is shown with the name of the task. The intervals are set in Service > Intervals. The factory settings correspond to the maintenance schedule in 4.1.

NOTE

Do not set longer intervals than in the maintenance schedule. Too long oil intervals cause varnish in the element and in the thermostatic valve.

3.5 Event log and service port

The event log stores the last 100 events. For every event the controller saves the date and time, the code, the running hours and the values of PT-1, PT-2, TS-1 and TS-2 at the moment of the event. The log can be read on the display or exported with the the VastraLink service software through the service port X10 on the controller board in the control compartment. Always attach the exported log to warranty claims and to questions to Vastra Technical Support.

3.6 Customer settings

Parameter	Description	Range	Factory
C1.01	Unload pressure	4.0 - max. working pressure	7.0 / 9.5 / 12.5 bar
C1.02	Load pressure	C1.01 minus 0.3 to 2.0 bar	C1.01 minus 0.8 bar
C1.03	Unloaded run time before stop	60 - 900 s	300 s
C1.05	Auto restart after power failure	Off / On	Off
C1.06	Restart delay after power return	10 - 600 s	30 s
C1.08	Control mode	Local / Remote / Modbus	Local

4 Maintenance

4.1 Maintenance schedule

The intervals are maximum intervals for normal operating conditions. The task is due at the interval in running hours or at the time interval, whichever comes first.

Interval	Task	Part no.	See
Daily	Control the oil level in the sight glass (compressor stopped for 2 minutes). Control the display for warnings.		4.2
500 h / monthly	Clean the pre-filter mats in the enclosure air inlet. Control the scavenge line sight glass for oil flow.		4.6
2,000 h	Replace the air filter element, or earlier at warning E-18.	41-1510	4.4
2,000 h	Clean the oil cooler and the aftercooler. Test the electronic water drain.		4.6
4,000 h / yearly	Change the oil.	41-0046	4.3
4,000 h / yearly	Replace the oil filter, or earlier at warning E-07.	41-1470	4.3
8,000 h	Replace the separator element.	41-3075	4.5
8,000 h / yearly	Test the safety valve. Control the minimum pressure valve and the coupling element.		4.7
24,000 h	Overhaul of the compression element bearings and shaft seal (Vastra service only).		

In dusty or humid environment, and when the ambient temperature is continuously above 40 C, the intervals for the air filter, the coolers and the oil shall be shortened. Contact Vastra service for advice on adapted intervals.

4.2 Oil

Vastra recommends Vastra SynLube 46 (synthetic, part no. 41-0046, 20 l can). It is allowed to use a mineral compressor oil ISO VG 46 according to DIN 51506 VDL in normal operating conditions. The oil change interval in 4.1 is the same for both oil types.

Do not mix oils of different types or makes. When changing from mineral oil to Vastra SynLube 46, drain the system completely, replace the oil filter, run the compressor for 24 hours and then change the oil and the oil filter a second time.

Control the oil level with the compressor stopped for 2 minutes. The level shall be in the upper half of the sight glass. Top up only with the same oil type. An oil consumption of more than 0.5 l per 100 running hours is not normal; see 5.2.4.

4.3 Changing the oil and the oil filter

1. Run the compressor until it is warm (discharge temperature above 70 C). Warm oil drains faster and takes more contamination with it.
2. Stop, isolate and vent the compressor according to 1.4. Control that PT-1 shows 0 bar.
3. Open the oil filler plug slowly by one turn. Remaining pressure escapes through the vent hole in the plug thread.
4. Connect a hose to the oil drain valve at the bottom of the separator vessel and drain the oil into a container. Drain also the oil cooler through its drain plug.
5. Remove the oil filter (41-1470) with a strap wrench. Clean the sealing surface on the filter head.
6. Oil the gasket of the new filter with clean oil. Screw the filter on by hand until the gasket touches, then turn a further 3/4 turn by hand. Do not use a tool.
7. Close the drain valve and the cooler drain plug. Fill oil through the filler opening until the level is in the upper half of the sight glass. Approximately 34 l is needed; the rest of the 38 l stays in the element and the pipes.
8. Fit the filler plug with a new O-ring. Run the compressor loaded for 5 minutes, stop, wait 2 minutes and control the level. Top up if necessary.
9. Reset the service counters for oil and oil filter in Service > Intervals. Write the date, the running hours and the oil type in the service record.

CAUTION

Used oil and used filters are hazardous waste. Collect them separately and hand them over to an authorised company.

4.4 Replacing the air filter element

1. Stop and isolate the compressor (1.4, steps 1 and 2 are sufficient).
2. Open the air filter cover clips and remove the cover. Pull out the element 41-1510 with a turning movement.
3. Clean the inside of the housing with a damp cloth. Do not blow into the inlet of the compressor.
4. Fit the new element and the cover. The arrow on the cover shall point upwards.
5. Reset the air filter service counter.

CAUTION

Never clean the paper element with compressed air. It damages the paper and lets dust into the element.

4.5 Replacing the separator element

Replace the separator element every 8,000 h. A clogged element increases the internal pressure and the energy consumption; a damaged element gives oil carry-over into the compressed air. The separator differential pressure can be followed in Measurements > Separator dP. Compare it with the value noted at the last replacement; a rising trend shows that the element comes to the end of its life.

1. Stop, isolate and vent the compressor (1.4). Control that PT-1 shows 0 bar.
2. Disconnect the scavenge line and the control line from the separator vessel cover.
3. Remove the cover bolts (M16) and lift off the cover with the lifting eye. Lift out the separator element 41-3075.
4. Clean the sealing surfaces. Control that the earthing staple in the new element is in place; it prevents static charge.
5. Fit the new element with the new gaskets from the kit. Fit the cover and tighten the bolts crosswise to 110 N-m.
6. Remove the restrictor from the scavenge line and clean the bore (1.5 mm) with a soft wire. Refit and reconnect the lines.
7. Run the compressor loaded and control that the oil flows in the scavenge line sight glass. Note the separator dP in the service record and reset the separator service counter.

4.6 Cleaning the coolers

Clean the oil cooler and the aftercooler from the fan side, against the normal direction of the cooling air, with compressed air at maximum 6 bar. Cover the motor and the electrical cabinet. Oily dirt is removed with a degreasing agent and water at low pressure. A steam cleaner or a high-pressure washer bends the fins and is not allowed. Clean also the pre-filter mats in the enclosure air inlet; they can be washed in warm water with a mild detergent.

4.7 Safety valve, minimum pressure valve and coupling

The safety valve shall be tested once per year, or according to the national regulations for pressure equipment if these are more strict. Lift the valve with the test lever with the compressor running unloaded. The valve shall blow and close tight. A valve that leaks after the test shall be replaced; it is not allowed to repair it.

Control that the minimum pressure valve opens and closes smoothly: at start, the net pressure (PT-2) shall begin to rise when the internal pressure (PT-1) has reached approximately 4 bar. Inspect the coupling element through the inspection hole for cracks and rubber dust.

5 Fault finding

5.1 Fault and warning codes

S = shutdown, the compressor stops. W = warning, the compressor continues to run. Codes that are not in the list are not used on the VRS-75. The measured values at the moment of the event are saved in the event log (3.5).

Code	Type	Display text	Condition	Action / see
E-01	S	Emergency stop	Emergency stop button pressed	Find out why it was pressed. Unlock the button, press Reset
E-02	S	Supply fault	Phase missing, wrong phase sequence or supply voltage outside +/-10%	Control the supply and the fuses. At first start, change two phases
E-03	S	Fan motor overload	Motor protection Q15 of the cooling fan	Control that the fan turns freely. Reset Q15
E-04	W	Internal pressure sensor	PT-1 signal outside 4 - 20 mA	Control the cable and plug at X12. Replace PT-1
E-05	S	Main motor overload	Thermal overload relay F21 tripped	Control the supply voltage and the motor current. Reset F21 by hand
E-06	W	Net pressure sensor	PT-2 signal outside 4 - 20 mA	Control the cable and plug at X12. Replace PT-2
E-07	W	Oil filter dP high	PS-3 closed, filter dP above 1.5 bar at operating temperature	Replace the oil filter (4.3)
E-09	S	Internal pressure low	PT-1 below 2.5 bar 20 s after load	5.2.2
E-10	W	Inlet temperature sensor	TS-1 open or short circuit	Control the cable and plug at X13
E-11	S	Net pressure high	PT-2 above max. working pressure + 0.3 bar	5.2.1
E-12	S	Low ambient temperature	TS-1 below +1 C. Start is blocked	Heat the compressor room. Do not start with frozen condensate
E-13	W	Discharge temperature	TS-2 above 105 C (pre-warning)	5.2.3
E-14	S	Discharge temperature high	TS-2 above 110 C	5.2.3
E-17	W	Drain alarm	Alarm contact of the electronic water drain	Clean the drain strainer. Control the drain valve
E-18	W	Air filter restriction	PS-4 vacuum switch closed (65 mbar)	Replace the air filter element (4.4)
E-19	W	Service due	A service interval counter has reached 0	Carry out the task shown, reset the counter (3.4)
E-20	S	Motor temperature	PTC thermistors B21 in the motor winding	Clean the motor fan cover. Control ambient and the number of starts (E-21)
E-21	W	Too many starts	More than 10 motor starts in the last hour	Control the pressure band (C1.01, C1.02) and the air demand
E-22	W	Remote communication	No remote signal or Modbus message for 10 s	Compressor runs in local mode. Control the remote connection

5.2 Fault finding procedures

5.2.1 E-11 Net pressure high

The net pressure has risen above the maximum working pressure although the compressor should unload at C1.01. Control in this order: that C1.01 is set correctly; that the inlet valve closes when the compressor unloads (listen for the change in sound; the internal pressure shall fall to approximately 2 bar); the loading solenoid valve Y1 and its coil; the PT-2 reading against a reference gauge at the air outlet. A compressor that stops on E-11 must not be restarted before the cause is found, because the safety valve is the only remaining protection.

5.2.2 E-09 Internal pressure low

The internal pressure does not reach 2.5 bar after loading, which means that the oil circulation is not assured. Causes: the inlet valve does not open (control Y1 and the inlet valve piston); the blow-off valve does not close; the minimum pressure valve is stuck open with an empty network; a large leak on the separator vessel. Control also the oil level; too little oil gives foaming and an unstable PT-1 reading.

5.2.3 E-13 and E-14 Discharge temperature

The controller stops the compressor when TS-2 measures more than 110 C at the element outlet. The pre-warning E-13 is shown at 105 C. In most cases the cause is insufficient cooling. Control in this order:

1. The temperature in the compressor room and the ventilation. The hot cooling air from the cooler outlet shall not come back to the cooling air inlet. With an outlet duct, control that the duct fan runs and that the duct is not blocked.
2. The coolers and the pre-filter mats. Clean them (4.6).
3. The oil level. Top up and look for oil leaks.
4. The cooling fan: it shall run when the motor runs, in the direction of the arrow on the fan cowl.
5. The thermostatic valve. If the oil cooler stays cold while the discharge temperature is high, the valve does not open. Replace the insert.
6. The oil filter (E-07 active?) and the age of the oil. An overdue oil change gives higher discharge temperature.

After an E-14 shutdown, let the compressor cool for at least 15 minutes before restart. If E-14 comes back and no cause is found, export the event log and contact Vastra Technical Support.

5.2.4 Oil in the compressed air

Oil in the compressed air, or a high oil consumption, has in most cases one of these causes: too much oil in the system after topping up; a blocked scavenge line restrictor (no oil flow in the sight glass); a damaged or old separator element; operation with long unloaded periods at low temperature, which gives foaming. Follow the separator differential pressure in Measurements; a sudden fall of the value can indicate a broken element.

5.2.5 The compressor does not start

Control the display for an active shutdown or a blocked start (E-12). In automatic mode the compressor waits for the load pressure (C1.02); the status shows "waiting". After a stop the minimum stop time of 30 s must pass before the motor can start again. In remote mode the start command must come from the remote contact or from Modbus (C1.08). After a power failure, the compressor only starts by itself if C1.05 is set to On.

6 Spare parts

6.1 Ordering

Always state the type, the serial number (data plate inside the electrical cabinet door) and the part number. Use only genuine Vastra parts and Vastra approved oils. The use of other parts can make the warranty invalid (7.1) and can damage the compressor. Spare parts that are not listed below are identified by serial number in Vastra PartsPortal, or through your Vastra service partner.

6.2 Service parts

Part no.	Description	Qty	Used in
41-0046	Vastra SynLube 46, synthetic compressor oil, 20 l can	2	4.3
41-1470	Oil filter, spin-on	1	4.3
41-1510	Air filter element (common with VRS-45)	1	4.4
41-3075	Oil separator element with gaskets (common with VRS-110)	1	4.5
41-2207	Discharge temperature sensor TS-2, PT1000, with plug	1	5.2.3

6.3 Service kits

Vastra service kits 2K (2,000 h), 4K (4,000 h) and 8K (8,000 h) contain the parts for the tasks of the maintenance schedule 4.1 at that interval. Oil is ordered separately. The kit numbers depend on the serial number; see Vastra PartsPortal.

6.4 Recommended stock

For a site with one or two VRS-75, Vastra recommends to keep on site one air filter element, one oil filter and 40 l of oil. Keep oil cans closed and stored indoors; opened cans shall be used within 6 months.

7 Warranty and documentation

7.1 Warranty

Vastra Kompressor AB gives a warranty for defects in material and manufacturing for 12 months from the date of commissioning, but at most 18 months from the date of delivery from the Vastra factory.

The warranty is valid on the following conditions:

- The commissioning report is sent to Vastra within 30 days from commissioning.
- Maintenance is carried out according to the maintenance schedule in this manual and is documented in the service record with date and running hours.
- Only genuine Vastra spare parts and Vastra approved oils are used.
- The compressor is installed and operated within the limits given in the technical data.

The warranty does not cover wear parts (filters, separator element, oil, coupling element), nor damage caused by wrong installation, insufficient ventilation of the compressor room, frost, condensate in the oil because of too low operating temperature, or the use of non-genuine parts. Warranty claims shall be sent by the authorised Vastra service partner within 30 days after the failure, with the serial number, the running hours and an export of the controller event log. Parts replaced under warranty are the property of Vastra and shall be kept for 60 days for a possible return.

7.2 Revision history

Rev.	Date	Changes
A	2021-09-06	First edition.
B	2024-02-12	3.4: service interval counters described. 4.5: separator element replacement revised, cover bolt torque changed to 110 N-m. 5.1: warnings E-21 and E-22 added. 6.2: air filter element 41-1510 now common with VRS-45.

7.3 Contact

Vastra Kompressor AB, Technical Support, Verkstadsgatan 9, SE-721 36 Vasteras, Sweden.

Technical Support is open Monday to Friday 07:00 - 16:30 CET. Outside Sweden, contact your authorised Vastra service partner first. Questions that the service partner cannot solve are handled by Vastra Technical Support (Level 2). Always state the serial number and the running hours.